

Linear Projector: Kernel and Image

1 Definition of a Linear Projector

A linear projector P is an idempotent operator satisfying:

$$P^2 = P$$

2 $\ker P$ (Kernel of P)

The kernel (null space) of P is the set of all vectors mapped to the zero vector:

$$\ker P = \{\mathbf{v} \mid P\mathbf{v} = \mathbf{0}\}$$

For a projector, this is the subspace of vectors "projected away" to nothing.

3 $\text{im } P$ (Image of P)

The image (range) of P is the set of all achievable outputs of P :

$$\text{im } P = \{\mathbf{w} \mid \exists \mathbf{v}, \mathbf{w} = P\mathbf{v}\}$$

This is the subspace that P projects *onto* — all valid results of the projection live here.

4 Key Decomposition

The full vector space V splits into a **direct sum** of these two subspaces:

$$V = \ker P \oplus \text{im } P$$

Every vector $\mathbf{v}$ has a unique decomposition:

$$\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$$

where $P\mathbf{v} \in \text{im } P$ and $\mathbf{v} - P\mathbf{v} \in \ker P$.

5 Example

Take the 2D projector:

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

- $\text{im } P = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ (the x-axis) - $\ker P = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ (the y-axis)

6 Isometries vs Unitary Operators

6.1 Definitions

- **Isometry** U : Linear operator with $U^*U = I$ (norm-preserving, injective).
- **Unitary operator** U : Linear operator with $U^*U = UU^* = I$ (norm-preserving, bijective).

6.2 Hierarchy

All unitary operators are isometries, but not all isometries are unitary (this distinction only appears in infinite-dimensional spaces). In finite dimensions, the two classes are identical.

6.3 Example: Right Shift on ℓ^2

The right shift operator R :

$$R(x_1, x_2, x_3, \dots) = (0, x_1, x_2, x_3, \dots)$$

is an isometry ($R^*R = I$) but not unitary ($RR^* \neq I$, as it is not surjective).